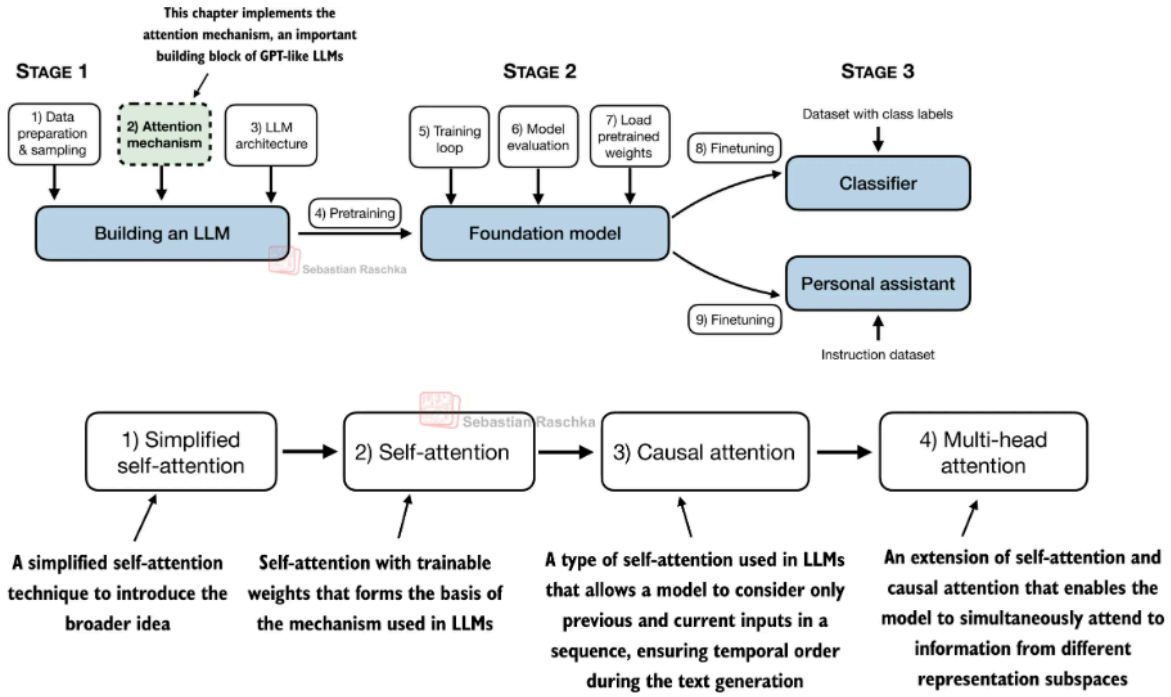
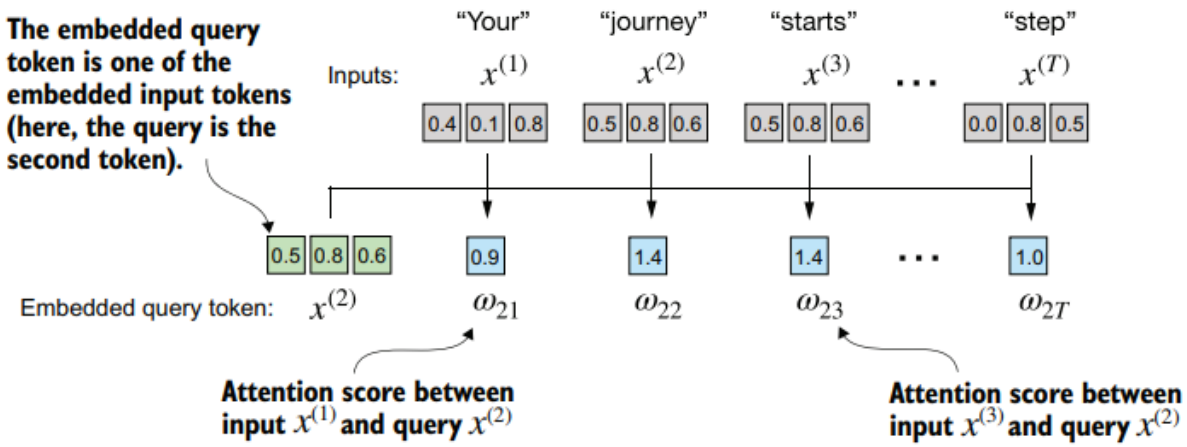


Gambar 1

1. A simple self-attention mechanism without trainable weights [ch03-Attention-Mechanism]



Gambar 2

Menghitung attention skor

```
[[0.43, 0.15, 0.89], # Your     (x^1)
 [0.55, 0.87, 0.66], # journey  (x^2)
 [0.57, 0.85, 0.64], # starts   (x^3)
 [0.22, 0.58, 0.33], # with     (x^4)
 [0.77, 0.25, 0.10], # one      (x^5)
 [0.05, 0.80, 0.55]] # step     (x^6)
```

Query token yang digunakan adalah x^2

Attention skor input x^1 (your) dan query x^2 (journey)

Dot product (w21) = (0.43 x 0.55) + (0.15 x 0.87) + (0.89 x 0.66)

= 0.2365+0.1305+0.5874

= 0.95544 ⇒ 0.9

Attention score antara x^1 dan x^2 adalah 0.95544 (sama seperti Gambar 2 dan Gambar 3)

```
# menghitung attention score
query = inputs[1]  # 2nd input token is the query

attn_scores_2 = torch.empty(inputs.shape[0])
for i, x_i in enumerate(inputs):
    attn_scores_2[i] = torch.dot(x_i, query)

print(attn_scores_2)
```

```
tensor([0.9544, 1.4950, 1.4754, 0.8434, 0.7070, 1.0865])
```

Gambar 3

#Menghitung Attention weight

Selanjutnya melakukan normalisasi pada **attention score** agar rentang angkanya antara 0 dan 1 dan jika dijumlahkan hasilnya 1.0, hasilnya disebut **attention weight**.

Pertama hitung total attention skor:

Total = 0.9544 + 1.4950 + 1.4754 + 0.8434 + 0.7070 + 1.0865 = 6.5617

Selanjutnya, bagi setiap attention skor dengan total

α21 = 0.9544 ÷ 6.5617 = 0.1455 (sesuai dengan attention weights Gambar 4)

α22 = 1.4950 ÷ 6.5617 = 0.2278

α23 = 1.4754 ÷ 6.5617 = 0.22485

α24 = 0.8434 ÷ 6.5617 = 0.1285

α25 = 0.7070 ÷ 6.5617 = 0.1077

α26 = 1.0865 ÷ 6.5617 = 0.16558

```
attn_weights_2_tmp = attn_scores_2 / attn_scores_2.sum()

print("Attention weights:", attn_weights_2_tmp)
print("Sum:", attn_weights_2_tmp.sum())
```

```
Attention weights: tensor([0.1455, 0.2278, 0.2249, 0.1285, 0.1077, 0.1656])
Sum: tensor(1.0000)
```

Gambar 4

attention weight dari hasil perhitungan menggunakan softmax ch03-softmax

α^{21} = 0.1385

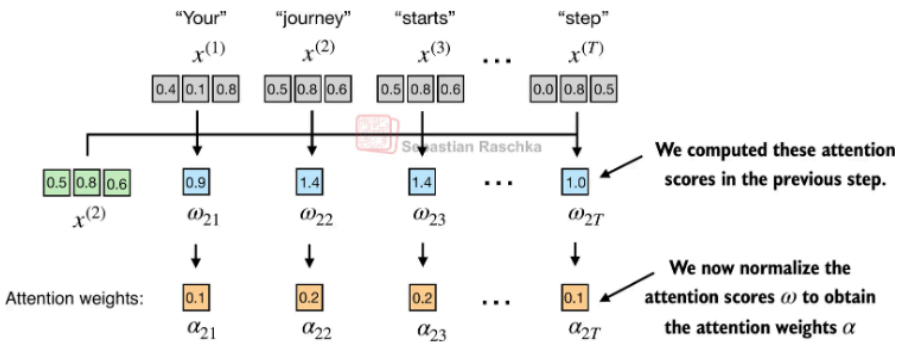
α^{22} = 0.2379

α^{23} = 0.2333

α^{24} = 0.1240

α^{25} = 0.1082

α^{26} = 0.1581



Menghitung context vector

Setiap input (x) dikali dengan attention weightnya (α) masing-masing, misal $x^1 \times \alpha^{21}$, kemudian semua hasilnya dijumlahkan untuk mendapatkan context vector.

⇒ input

```
[[0.43, 0.15, 0.89], # Your     (x^1)
 [0.55, 0.87, 0.66], # journey  (x^2)
 [0.57, 0.85, 0.64], # starts   (x^3)
 [0.22, 0.58, 0.33], # with     (x^4)
 [0.77, 0.25, 0.10], # one      (x^5)
 [0.05, 0.80, 0.55]] # step     (x^6)
```

hitung context vector untuk elemen pertama (indeks 0):

(Kalikan setiap elemen pertama dari vektor input dengan bobotnya lalu jumlahkan)

$z^{(2)} = (x^1 \times \alpha^{21})+(x^2 \times \alpha^{22})+(x^3 \times \alpha^{23})+(x^4 \times \alpha^{24})+(x^5 \times \alpha^{25})+(x^6 \times \alpha^{26})$

$z^{(2)}$ = (0.43×0.1385)+(0.55×0.2379)+(0.57×0.2333)+(0.22×0.1240)+(0.77×0.1082)+(0.05×0.1581)

$z^{(2)}$ = 0.059555 + 0.130845 + 0.132981 + 0.027280 + 0.083314 + 0.007905

$z^{(2)}$ = 0.059555 + 0.130845 + 0.132981 + 0.027280 + 0.083314 + 0.007905

$z^{(2)}$ = 0.44188

$z^{(2)}$ = 0.44188 (0.4419)

Untuk elemen kedua (indeks 1):

=(0.1385×0.15)+(0.2379×0.87)+(0.2333×0.85)+(0.1240×0.58)+(0.1082×0.25)+(0.1581×0.80)

=0.020775+0.206973+0.198305+0.071920+0.027050+0.126480

=0.020775+0.206973+0.198305+0.071920+0.027050+0.126480

=0.651503

=0.651503 (0.6515)

Untuk elemen ketiga (indeks 2):

=(0.1385×0.89)+(0.2379×0.66)+(0.2333×0.64)+(0.1240×0.33)+(0.1082×0.10)+(0.1581×0.55)

=0.123265+0.157014+0.149312+0.040920+0.010820+0.086955

=0.123265+0.157014+0.149312+0.040920+0.010820+0.086955

=0.568286

=0.568286 (0.5683)

